Dirichlet

<u>Def</u>: $\{a_n\}_{n\geq 1} \xleftrightarrow{Dir} f(s) = \sum_{n\geq 1} \frac{a_n}{n^s}$

$\{1\}_{n\geq 1} \xleftrightarrow{Dir} \zeta(s)$

<u>Fact</u>: $\{a_n\}_{n\geq 1} \xleftrightarrow{Dir} f(s)$

$\{b_n\}_{n\geq 1} \xleftrightarrow{Dir} g(s)$

$k, l \in \mathbb{R}$, $\{ka_n + lb_n\} \xleftrightarrow{Dir} kf(s) + lg(s)$

<u>Thm</u>: $\{\sum_{d|n} a_d b_{\frac{n}{d}}\}_{n\geq 1} \xleftrightarrow{Dir} f(s)g(s)$

$\{\sum_{ijk=n} a_i b_j c_k\}_{n\geq 1} \longleftrightarrow f(s)g(s)h(s)$

<u>Eg</u>: $\{d(n)\} \longleftrightarrow \zeta^2(s)$

$\downarrow$

# of positive divisor of n

<u>Def</u>: f is <u>number-theoretic</u> if its domain is $\mathbb{Z}^+$

f is multiplicative 积性. if

$f(mn) = f(m)f(n)$ if $\gcd(m,n)=1$

For example $d(n)$ (as above)

Thm: f is multiplicative $\Rightarrow \{f(n)\}_{n\geq 1} \xleftrightarrow{\text{Dir}} \prod_{p:\text{ all primes}} \sum_{k\geq 0} \frac{f(p^k)}{p^{ks}}$

Eg: Mobius function

$\mu \quad \mathbb{Z}^+ \longrightarrow \mathbb{Z}$ is multiplicative

$$n \mapsto \begin{cases} 1 & n=1 \\ -1 & n=\text{prime} \\ 0 & n=p^a \quad a\geq 2 \end{cases}$$

Eg: $f(n)\equiv 1$

$$\zeta(s) = \prod_{p:\text{prime}} \left(\sum_{k\geq 0} p^{-ks}\right) = \prod_p \frac{1}{1-p^{-s}} \qquad \mu(n)$$

Thm (Inclusion exclusion. classical